



BSc (Hons) in **Information Technology**
Specialized in AI
IT3012 : Intelligent Agents

Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 02

Objective & Lecture Mapping: In Lecture 03, we learned that a mathematically perfect "Table-Driven Agent" is impossible to build due to combinatorial explosion. Instead, we must write Agent Programs. Today, you will implement a **Simple Reflex Agent** using Condition-Action rules, observe its fatal flaws in a partially observable environment, and then upgrade it to a **Model-Based Agent** that maintains an internal memory state.

Part 1: Practical Implementation — Coding the Architectures

Step 1.1: Setting the Trap (Partial Observability)

To see why Simple Reflex agents fail, we must handicap your agent's sensors. We are moving from a Fully Observable world to a Partially Observable one.

1. Open your `visual_grid_game.py` from Lab 01.
2. Locate the `get_percept(self)` method.
3. Modify it so it **no longer returns** the exact global coordinates (`agent_pos`). Instead, it should only return local booleans, e.g., `{ 'wall_ahead': True/False, 'food_here': True/False }` based on checking the adjacent cell in its current facing direction.

Step 1.2: The Simple Reflex Agent (Implementation & Failure)

1. Create a new class called `SimpleReflexAgent` .
2. Implement a `sense_and_act(self, percept)` method that uses strictly IF-THEN logic (Condition-Action rules). *Do not use an `__init__` method to store history.*

Example Logic: IF food_here THEN suck; IF wall_ahead THEN turn_left; ELSE move_forward

3. Run the simulation. **Observe:** Watch the agent get trapped in a corner or a U-shaped wall, infinitely repeating a cycle (e.g., turn left, step forward, hit wall, turn left...).

Step 1.3: The Model-Based Agent (Memory & State)

1. Create a new class called `ModelBasedAgent`.
2. Implement an `__init__(self)` method to initialize an internal memory state (e.g., `self.visited_cells = set()` or a relative movement tracker).
3. In `sense_and_act(self, percept)`, first **update the state** (Transition & Sensor Model) by recording the current percept and the last action taken.
4. Update your IF-THEN rules to query the memory.
Example Logic: IF wall_ahead AND left_is_visited THEN turn_right
5. Run the simulation. **Observe:** The agent should now remember where it has been, realize it is in a loop, and choose an alternate path to escape.

Part 2: Theoretical Evaluation & Lecture Mapping

Based on your practical observations above, answer the following questions to map your code directly to the architectural theories covered in Lecture 02.

1. (Remember) According to Lecture 02, why is it impossible to program a mathematically perfect "Table-Driven Agent" for complex environments like Chess? What happens as the agent's lifetime increases?

A mathematically perfect table-driven agent is impossible for complex environments like Chess because the size of the percept history table expands exponentially through combinatorial explosion, quickly surpassing the total number of atoms in the observable universe. As an agent's lifetime increases, the required lookup table grows infinitely, rendering storage, precomputation, and retrieval completely infeasible.

2. (Understand) Look at the code you wrote for your `SimpleReflexAgent`. Identify and explain the specific lines of code that represent the "Condition-Action Rules"

discussed in the lecture.

Condition-action rules in code are explicitly represented by conditional branches (such as if-elif-else blocks) inside the agent's evaluation method. These statements directly map a specific sensory input (e.g., if `percept['wall_ahead']`: return 'Left') to a motor action without evaluating past history or maintaining internal state variables.

3. (Analyze) Your `SimpleReflexAgent` likely got stuck in an infinite loop during Step 1.2. Based on the lecture, analyze exactly why this happened. How did the combination of "Partial Observability" and a lack of "Percept History" cause this failure?

The infinite loop occurs because partial observability restricts the agent's view to immediate, local surroundings rather than global map coordinates. Combined with a lack of percept history or memory, encountering the same local environment (like a corner) produces identical sensory inputs, triggering the exact same reflex action repeatedly. The agent cannot recognize repetition because it lacks the temporal

4. (Evaluate) In Step 1.3, you added an internal state to your `ModelBasedAgent`. Evaluate how your specific code handles the "Transition Model" (how the world evolves) and the "Sensor Model" (how the agent's actions affect the world).

Sensor Model: The agent interprets incoming local percepts to update its internal representation of the world, bridging the gap caused by partial observability by mapping current sensory data to hidden features of the environment.

Transition Model: The internal update logic tracks how actions alter the world and the agent's position (e.g., updating a `visited_cells` set or relative coordinate tracker using the last action taken), allowing the agent to predict state changes and choose alternative paths when trapped.

Finalize and Lock Lab 02 Documentation

Incomplete: 0/11 questions answered. Please provide detailed responses for all questions.



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